Автоматический синтез речи. Статистический параметрический синтез

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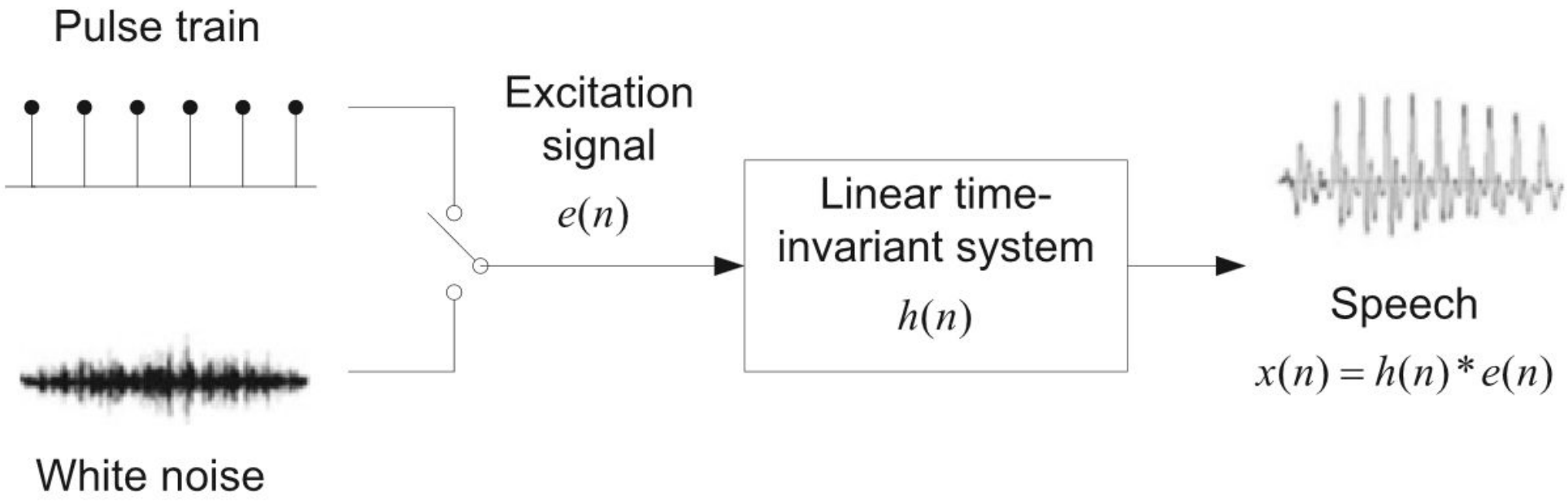
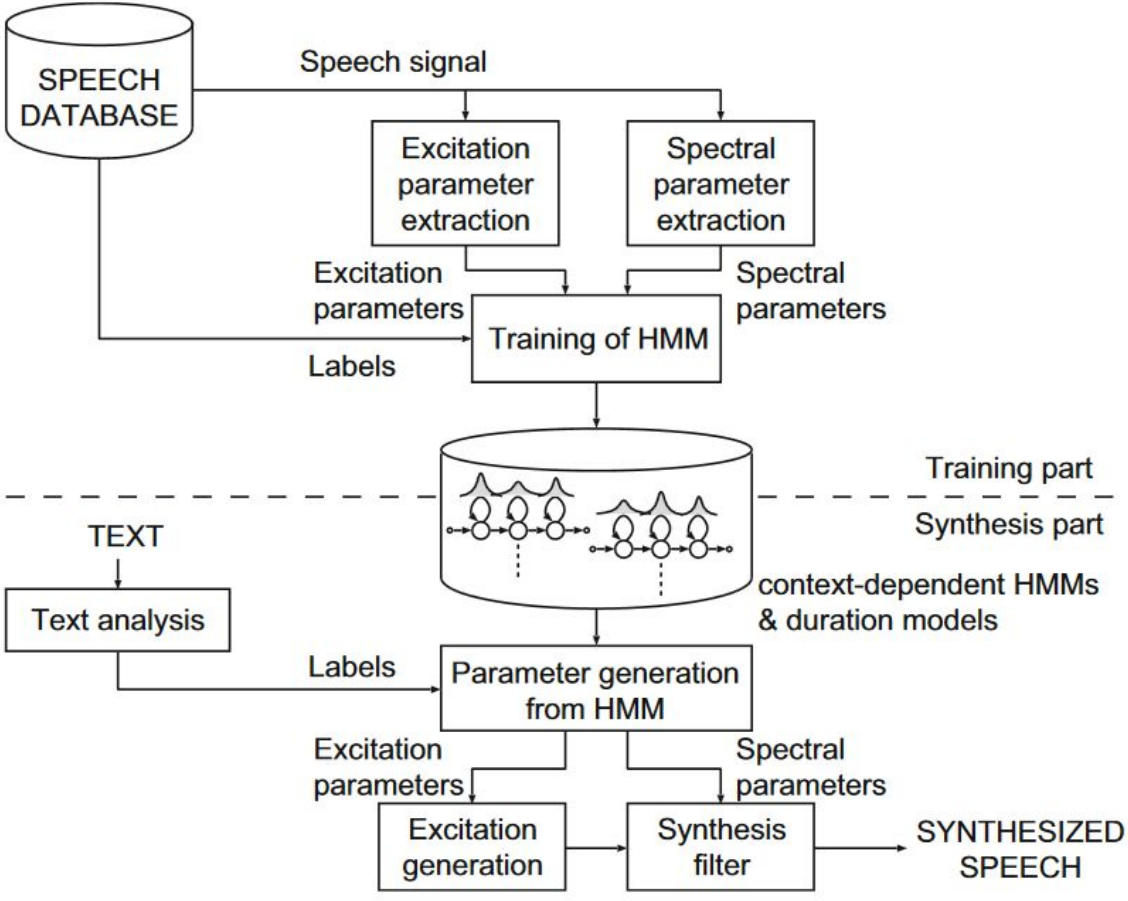
Модель источник-фильтр



Скрытые Марковские модели



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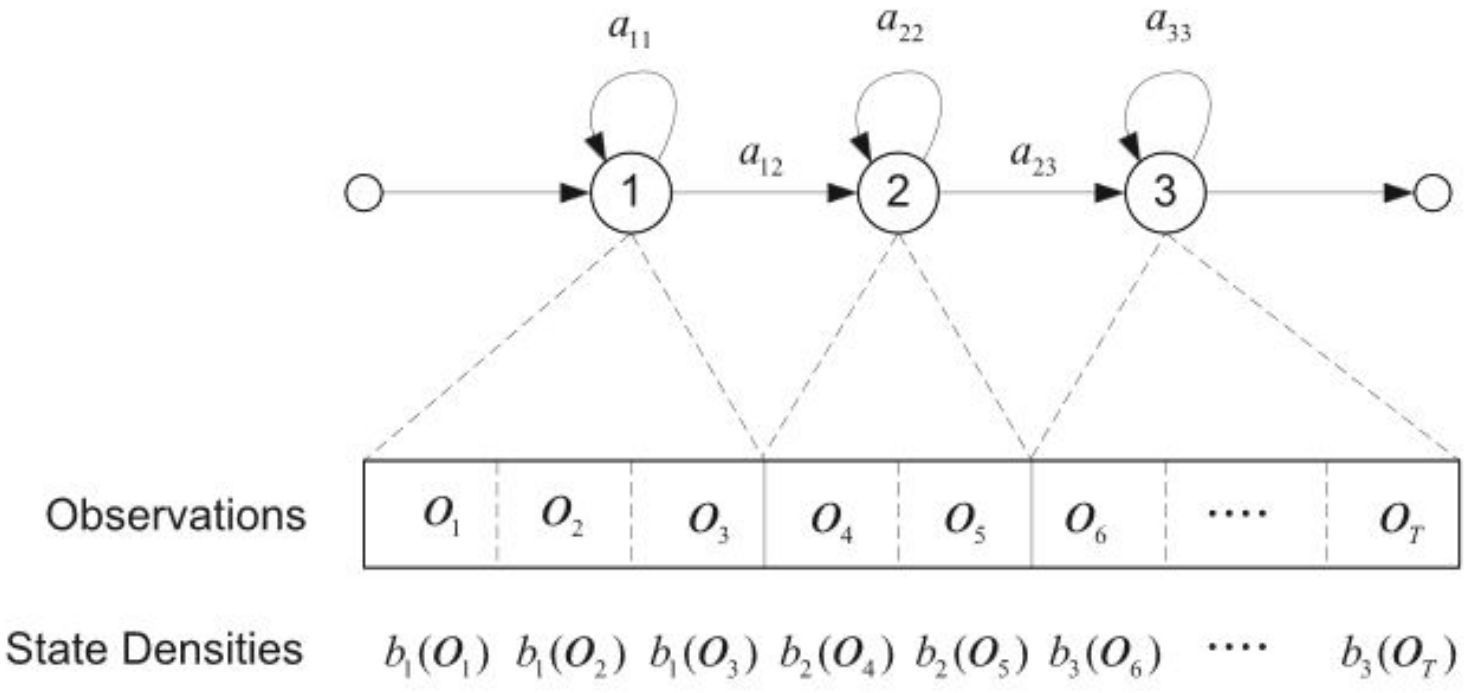


Скрытые Марковские модели

1. Формально:

2. Обучение:

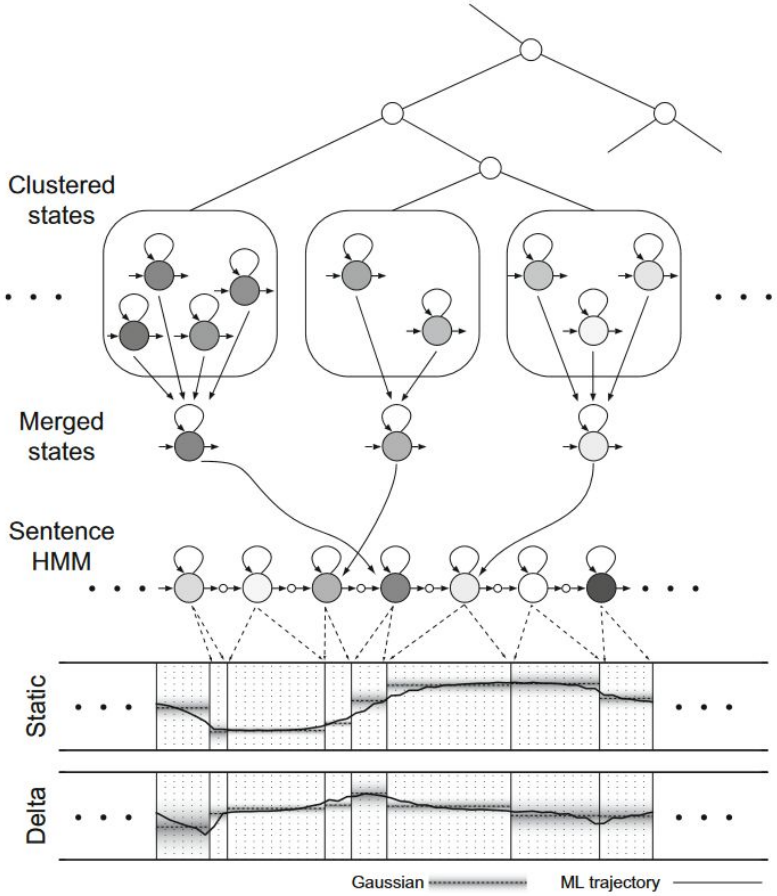
$$\hat{\boldsymbol{\lambda}} = \arg\max_{\boldsymbol{\lambda}} \{p(\boldsymbol{O}|\mathscr{W}, \boldsymbol{\lambda})\}$$

3. Синтез:

$$\hat{\boldsymbol{o}} = \arg\max_{\boldsymbol{o}} \left\{p(\boldsymbol{o}|w, \hat{\boldsymbol{\lambda}})\right\}$$

Скрытые Марковские модели

1. Проблема: генерируемый сигнал должен быть гладким



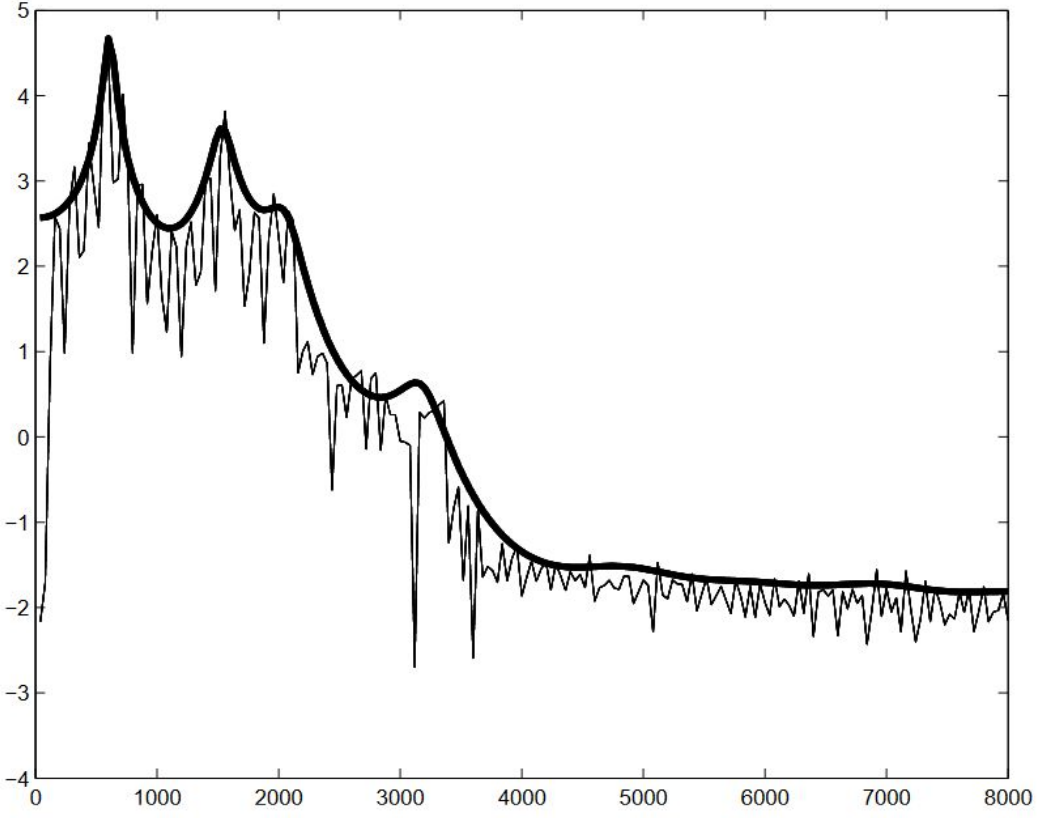
Линейное предсказание

$$y[n] = x[n] \otimes h[n]$$

$$\begin{aligned} Y(z) &= X(z)H(z) \\ &= \frac{1}{1 - \sum_{k=1}^{P} a_k z^{-k}} X(z) \end{aligned}$$

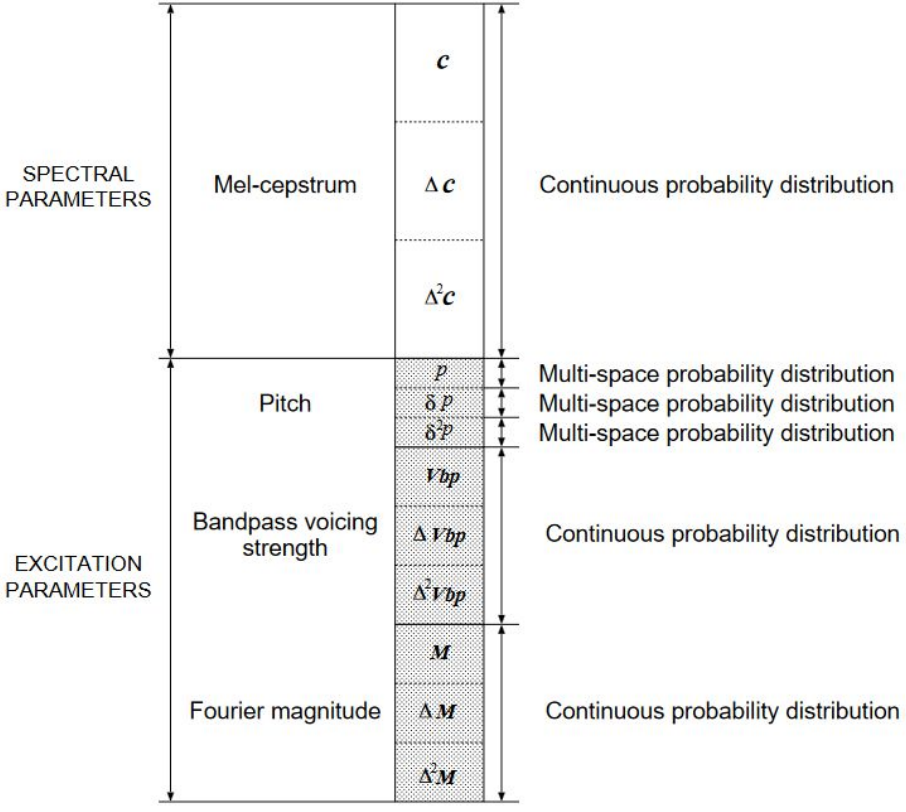
$$y[n] = \sum_{k=1}^{P} a_k y[n-k] + x[n] \longrightarrow \tilde{y}[n] = \sum_{k=1}^{P} a_k y[n-k]$$

Линейное предсказание



Моделирование источника

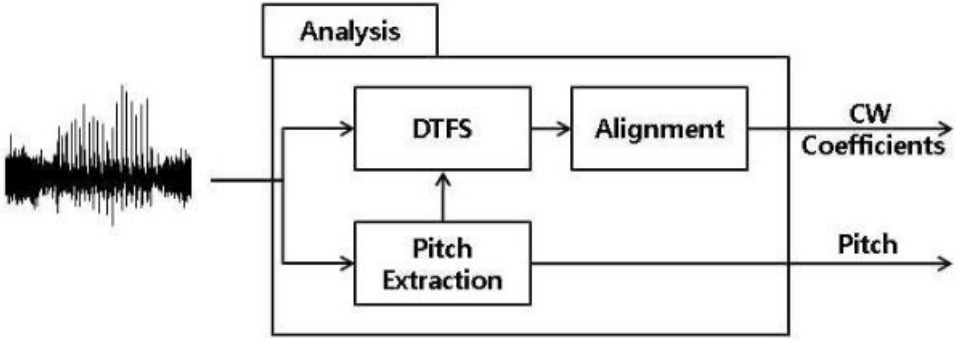
Mixed excitation

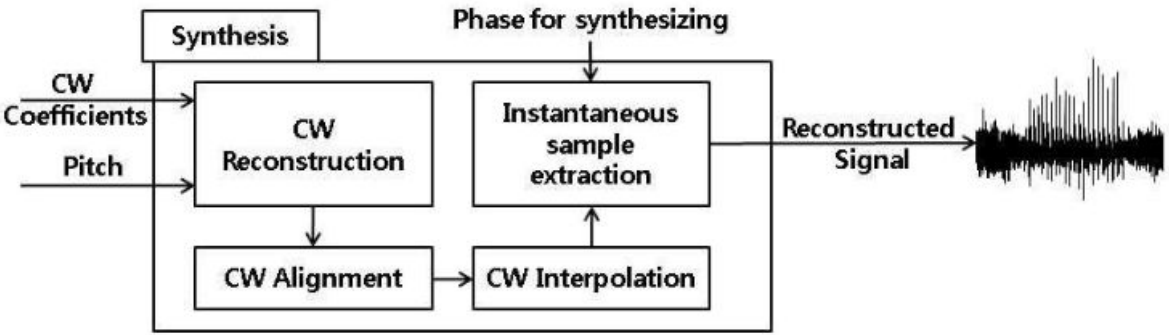


Моделирование источника

Waveform interpolation:

CW — characteristic waveform





Моделирование источника

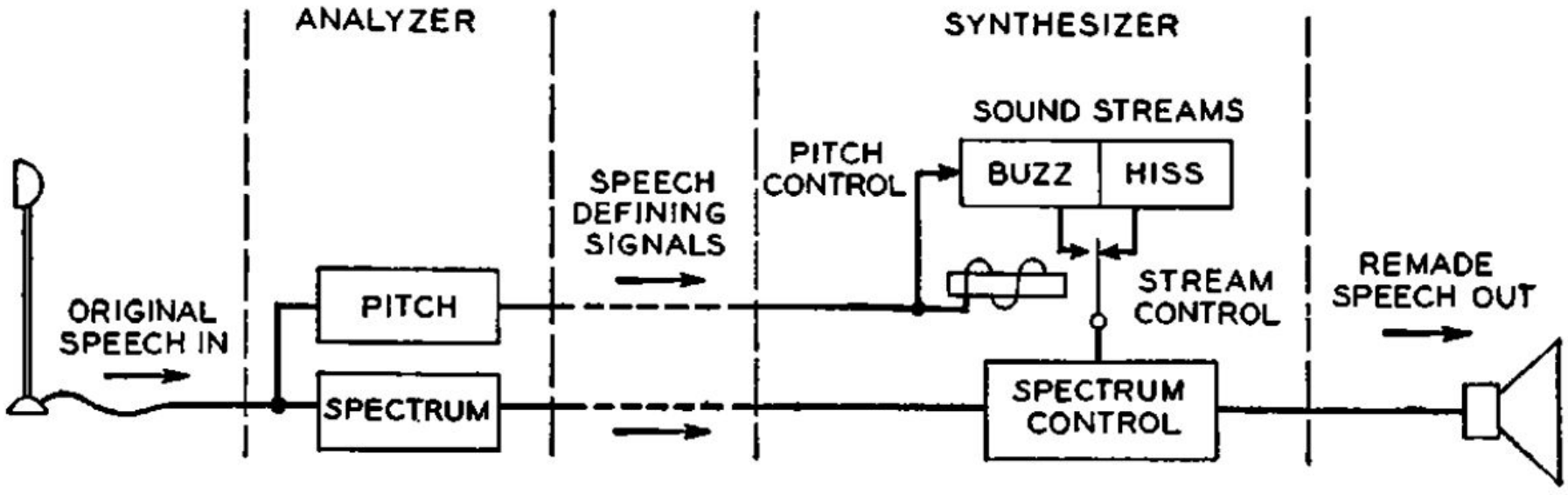
Liljencrants-Fant (LF) model

$$e_{LF}(t) = \begin{cases} E_0 e^{\alpha t} \sin(w_g t), & t_o \leq t \leq t_e \\ -\frac{E_e}{\epsilon T_a}[e^{-\epsilon(t-t_e)} - e^{-\epsilon(t_c-t_e)}], & t_e < t \leq t_c \\ 0, & t_c < t \leq T_0 \end{cases}$$

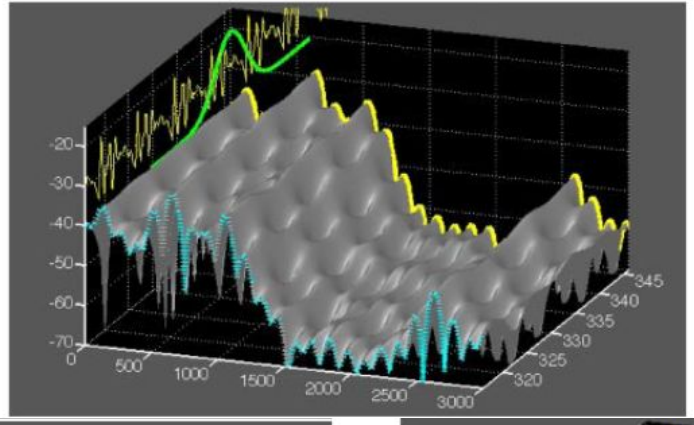
Моделирование фильтра

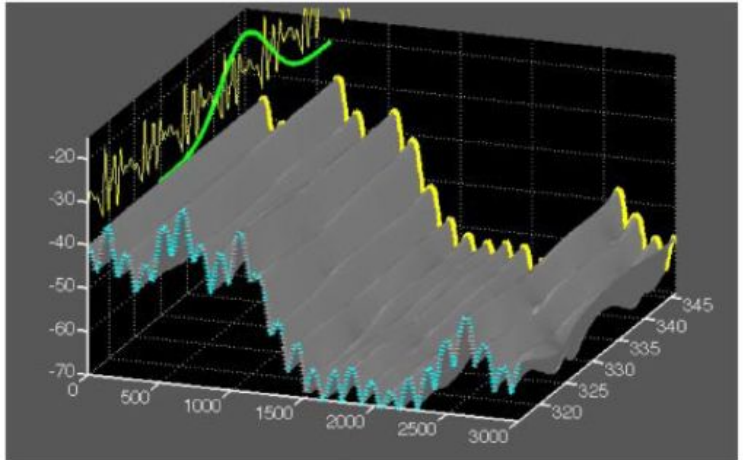
1. Мел-кепстральные коэффициенты (MFCC)
2. Мел-спектрограммы
3. Барк-кепстральные коэффициенты
4. Band aperiodicity

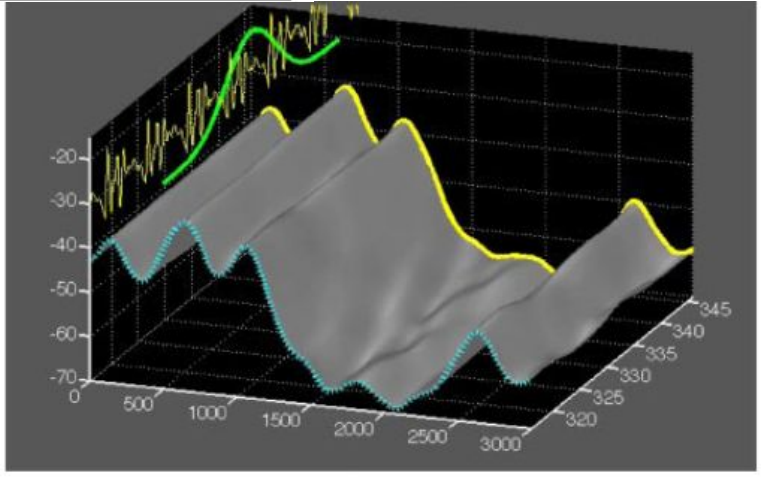
Вокодеры: VOCODER



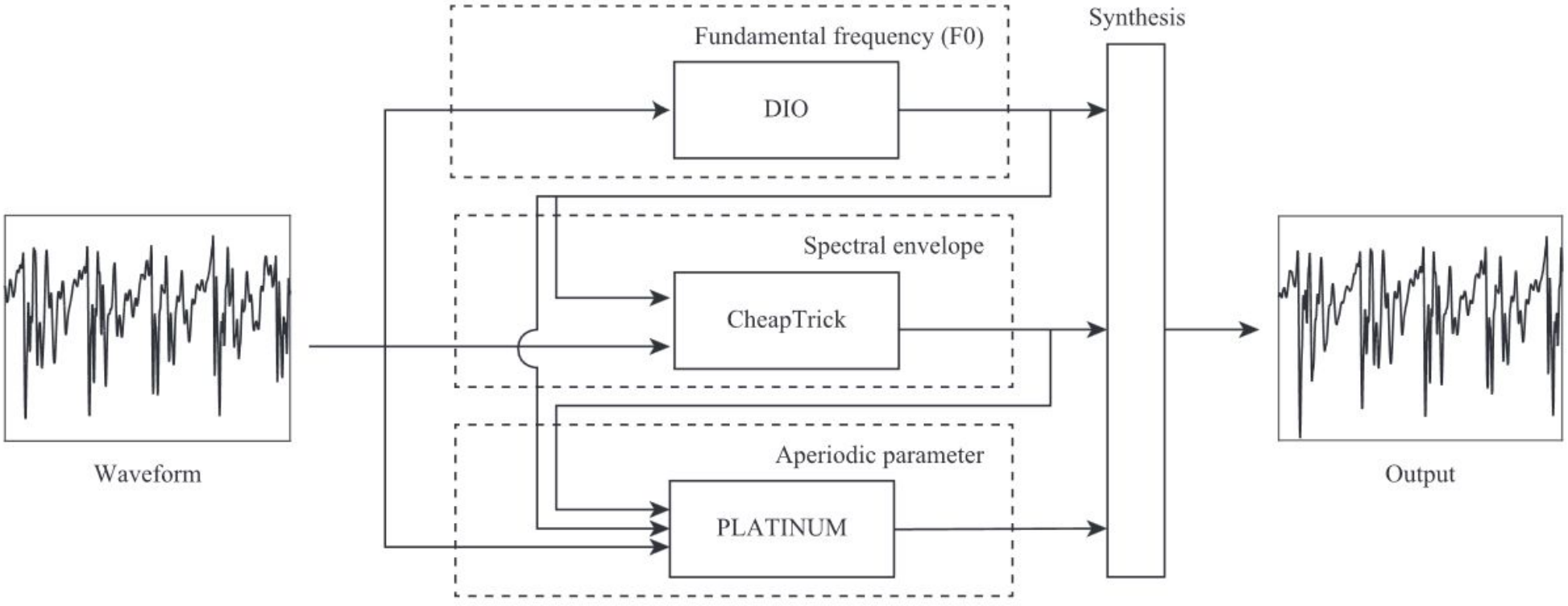
Вокодеры: STRAIGHT



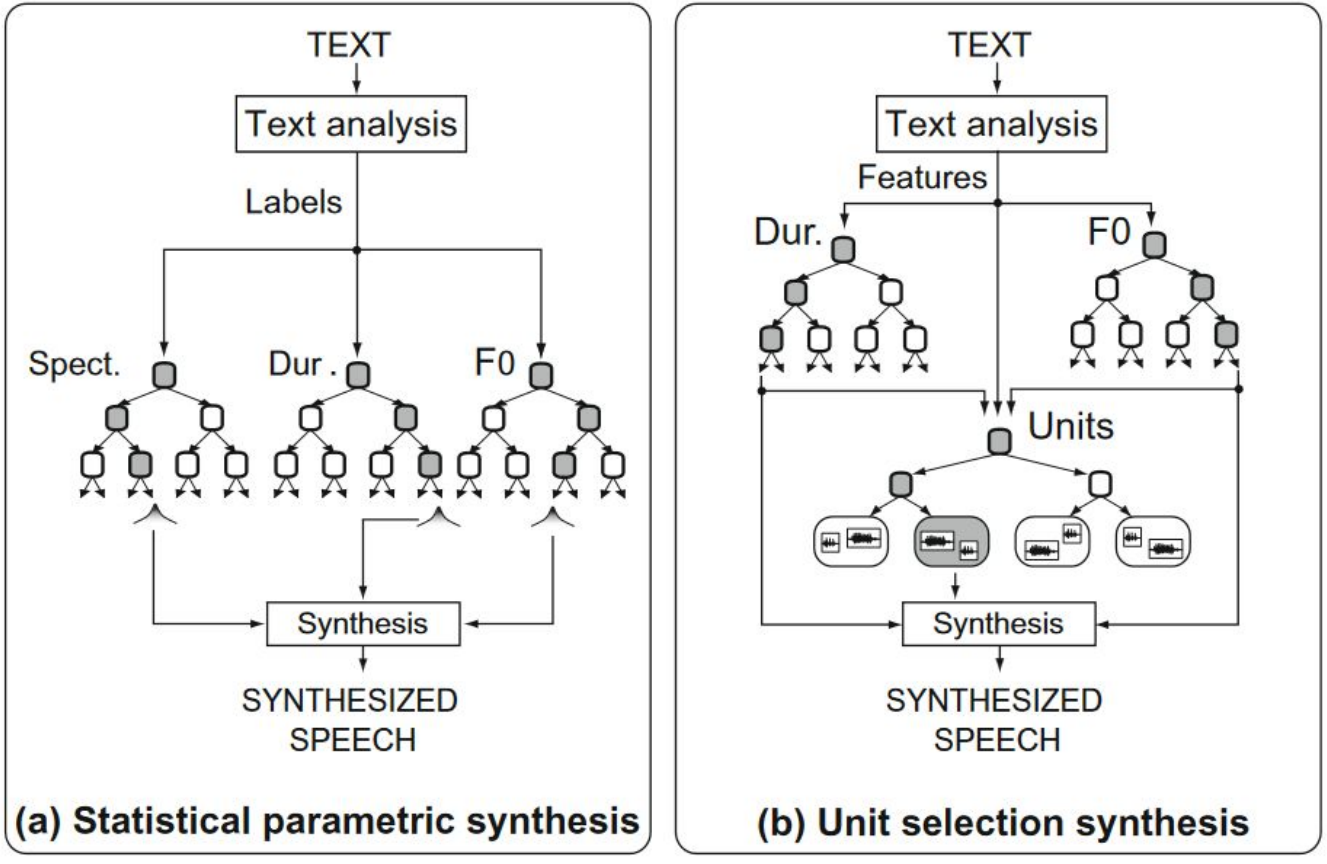




Вокодеры: WORLD



Деревья принятия решений



Спасибо за внимание!